

Regression Analysis (I)

Final Exam.

June 14, 2021

1.(250) Consider a multiple linear regression model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim N_n(\mathbf{0}, \mathbf{I}\sigma^2)$, where $\mathbf{X}$ is $n \times p$ ($p < n$) full column rank matrix. Let $\hat{\boldsymbol{\beta}}$ be the LSE of $\boldsymbol{\beta}$, $\mathbf{H} = (h_{ij})$ be the hat matrix, and $\mathbf{e} = (e_1, \dots, e_n)^T$ be the residual vector.

(1)(50) Show that $Cov(\hat{\boldsymbol{\beta}}^*) - Cov(\hat{\boldsymbol{\beta}})$ is non-negative definite, where $\hat{\boldsymbol{\beta}}^* = [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}]\mathbf{y}$ is an unbiased estimator of $\boldsymbol{\beta}$ for some $p \times n$ matrix $\mathbf{D}$ with rank p .

(2)(50) Let $\hat{\boldsymbol{\beta}}_{(i)}$ be the LSE of $\boldsymbol{\beta}$ based on $(n - 1)$ observations after deleting the i th observation. Using $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i e_i / (1 - h_{ii})$, show that the i th Cook's distance can be written as a function of the residual and the leverage.

(3)(50) Derive the distribution of $(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^T \mathbf{X}^T \mathbf{X} (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) / ps^2$, where $s^2 = \sum_{i=1}^n e_i^2 / (n - p)$.

(4)(50) Derive the distribution of $e_i / s_{(i)} \sqrt{1 - h_{ii}}$, where $s_{(i)}^2 = \sum_{j \neq i}^n (y_j - \mathbf{x}_j^T \hat{\boldsymbol{\beta}}_{(i)})^2 / (n - p - 1)$.

(5)(50) Find LSE of $\boldsymbol{\beta}$ under the restriction of $\boldsymbol{\beta}^T \boldsymbol{\beta} = c$ for some $c > 0$.

2.(100) Consider an AR(1) model, i.e, two adjacent error terms are correlated as $\epsilon_t = \rho\epsilon_{t-1} + \delta_t$, $t = 0, 1, \dots$, where $E(\epsilon_t) = 0$ and $Var(\epsilon_t) = \sigma^2$. Assume that $|\rho| < 1$ and δ_t 's are iid r.v.'s with $E(\delta_t) = 0$ and $Var(\delta_t) = \gamma^2$.

(1)(50) Show that $E(\epsilon_t - \epsilon_{t-1})^2 = 2\sigma^2 - 2\rho\gamma^2 / (1 - \rho^2)$

(2)(50) Derive a Durbin-Watson test statistic using the result in (1).

3.(50) Define the Kullback-Leibler (KL) distance. When does the KL distance achieve minimum value?